Meta & Multi-Task Learning Project 3

(For Graduate Students)

Due Date: 12/21 23:59

Overview

In this project, your objective is to build efficient, lightweight few-shot learning algorithm that surpasses old baselines on *mini*-ImageNet under budget constraints.

Codes [80pt (+ 15pt)]

- *mini*-ImageNet
  - Performance [70pt (+ 15pt)]
  - Ablation [10pt]

Report [20pt (+5pt)]

- In-depth analysis on chosen method(s) and ablation study.

Codes

Your goal is to select few-shot/meta-learning learning algorithm(s) introduced in this course, and overwhelm the previous baseline methods (MAML, prototypical network with Conv4) in terms of accuracy and speed on *mini*-ImageNet benchmark (5-way 1-shot.)

Unlike previous projects, hard constraints barely exist in this project. You may refer to any public codes, or even import external meta-learning or few-shot learning libraries. Yet, the references should be clarified in the report.

Action Budgets

Here are actions that will lead to **zero** score:

- Using test split in training.
- Downloading external/pretrained model weights or extra datasets.
- Training settings other than inductive supervised learning (e.g. transductive, semi-supervised)

Here are actions with soft constraints (the penalties are *additive* to your final score):

- Referring to public codes (or copy-pasting): ok
- Training configuration adjustment (optimizer/scheduler, epoch, backbone, etc.): ok
- Importing external libraries (requiring extra installation): -10pt
- Introducing new meta-learning/few-shot learning algorithm(s): -10pt
- Revealing ground truth label of training data: -10pt
- Augmentations, ensembles (MoE), AutoML, calibration: ok without external library.

Example action plans:

- Changing backbone network of prototypical networks with automated hyperparameter selections for optimizer/scheduler/network with *external* AutoML libraries.
    - AutoML libraries included: -10pt
    - Applying the selected hyperparameter and discarding the library: ok (sidestepping penalty.)
- LEO with *ground-truth label* in embedder pretraining.
    - Training from scratch: -10pt
    - Downloading the *pretrained* weights: zero score.
- R2D2 + protonet + SAM + transforms.v2.RangAugment + MoE soft gates, train from scratch: ok
- Bringing unlearned SOTA few-shot learning algorithms from class
    - No external library, no ground-truth label used, no pretrained weights, inductive: -10pt
    - External library, ground-truth label, no pretrained weight, inductive: -30pt
    - Utilizing *pretrained* weight or *external* data: zero score
    - *Transductive* algorithm: zero score

A checklist on your action plan will be provided in the report template – don't be overwhelmed to calculate.

Code Budgets

There are three components for performance evaluation – accuracy score (a), training time score (t), and parameter score (p). The final performance score will be calculated by multiplication of three factors ($a \cdot t \cdot p$).

Accuracy score $a(x) = max\left\{1.2 \cdot \left(\frac{0.5(x-50)}{1+|0.5(x-50)|}\right),\ 0\right\}$

- Your code should surpass 50% accuracy on *mini*-ImageNet 5-way 1-shot experiment.
- {55%, 60%, 65%, 70%} accuracy: {0.86, 1.00, 1.06, 1.09}

Training time score $t(x) = 0.5 \cdot \tanh(-x + 7) + 0.6$ (Hard cutoff: 9 hours)

- < 5.9 hours to get full credit or more.
- {4, 5, 6, 7} hours: {1.10, 1.08, 0.98, 0.6}

Parameter score $p(x) = 0.5 \cdot \tanh(-x/10M + 3) + 0.55$ (Hard cutoff: 50M)

- < 15.2M total parameters to get full credit or more.
- {5M, 10M, 20M, 30M} total parameters: {1.04, 1.03, 0.93, 0.55}

Final performance score: $\mathbf{70 \cdot a \cdot t \cdot p} \in [0,\ 92.98)$

- If your final performance score surpasses the maximum point (70pt), it will become your bonus point for the overall project scores.
- Practically, you can get roughly up to ~15 bonus points at maximum.

You should also conduct *ablation* experiment. The ablation study must include the most important component of your approach to improve model accuracy or(and) training time.

Report

Unlike previous projects, Latex template is provided (ICML style):

https://www.overleaf.com/read/qwqfktzdwyqd#1849a1

The detailed instructions are in the template. Please fill out the checklist and attach in your final submission **(mandatory)**.

Submission

Submit a single zip file ('proj3_<student_ID_number>.zip') that contains all the running codes of your implementation, as well as the report .pdf file. Hierarchies do not matter, but do not include the *mini*-Imagenet dataset itself.